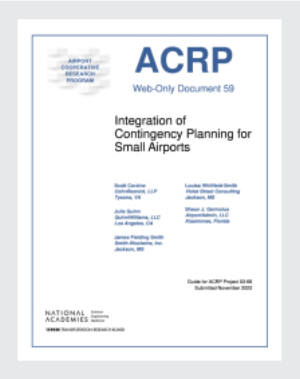
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ACRP

Web-Only Document 59

Integration of Contingency Planning for Small Airports

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Guide for ACRP Project 03-68
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Guide Documentation

Overview

Small airports[1] are tasked with meeting the same resiliency requirement as larger airports, while operating under the burden of many constraints, including limited personnel and resources. For this reason, contingency planning is crucial so that small airports can best position themselves to be prepared to address disruptions and contingencies - any situations that impact staffing, financial and information technology resources, infrastructure, facilities, and/or supply chain shortages.

Plan integration may be conceived of as a daunting task requiring a level of stakeholder engagement and the deployment of scarce resources out of reach for many small airports where operations, particularly during the COVID-19 pandemic, were stretched thin and airport staff were forced to "wear ever more hats". Yet COVID-19 also revealed the necessity of effective plan integration for operational and business continuity, emergency response, and resiliency during airport disruptions.

Plan integration is also important as airports prepare to respond to regional events as well as on-site incidents. The airport might be the primary site of the incident or serve as a resource during municipal or county response to a wider-scale scenario. Larger incidents require "whole of area" integration at both planning and response/recovery stages. Such events also necessitate appropriate and equitable resource allocation, be they food, medical, equipment, or shelter related.

There is a greater need for contingency plan integration when physical and capital resources are being shared. This is especially the case for small airports that rely heavily on city/county resources. Integration tools therefore need to also factor in the governance structure that the airport must work within, whether city/county government or airport authority. Tools that can integrate airport contingency plans with the plans of sponsors and neighboring jurisdictions and capabilities are likely to be most useful during real-world events.

Plan Integration Tool Kit

The stated objective of ACRP 03-68 is to provide resources, tools and examples of integration that can guide future efforts undertaken by small airports. Smart, integrated contingency planning can optimize the functionality of an airport during a disruption or in a local or regional disaster and minimize the operational and economic impacts to the airport and to its many stakeholders and "dependents."

1 The term "small airports" is inclusive of general aviation (GA), secondary/commercial service (CS) airports, non-hub primary (NH-P) airports, and small-hub (SH) airports as defined in the National Plan of Integrated Airport Systems (NPIAS) and FAA annual enplanement data.

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The ACRP 03-68 research team created and/or identified nine (9) integration tools sorted into three (3) categories:

- Plans and Planning Tools
- Implementation Tools
- Communications Tools

Our research—particularly the interviews with the airports--revealed that our development objectives ensure that the tools are:

- Practical and effective for small airports
- Easy to use
- Not administratively burdensome
- Operationally focused
- Easy to access in hard copy format
- Free

Role of Threat and Hazard Identification and Risk Assessment (THIRA)

Small airports may want to consider using a Threat and Hazard Identification and Risk Assessment (THIRA) to help identify and prioritize the plans you'll be considering in this toolkit. THIRAs are a three-step risk assessment process that helps the airport understand your risks and what you need to do to address those risks by addressing:

- What threats and hazards can affect the airport?
- What impacts would those threats and hazards have on the airport if they occurred?
- Based on those impacts, what capabilities should we have in place and what plans should we prioritize?

If your airport has not completed a THIRA, consider an inexpensive abbreviated risk assessment that addresses your particular risk circumstances and how well your staff and resources are capable of responding to the risk impacts you may face.[2] Another alternative is to work with the airport's local, regional, or state emergency management or planning agency to develop a full THIRA.

In each of the nine tools, there are references to other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans.

The nine tools are provided herein. Interactive versions of each tool are also available as individual PDF downloads on the National Academies Press website (nap.nationalacademies.org) by searching for *ACRP Web-Only Document 59: Integration of Contingency Planning for Small Airports*.

2 https://www.fema.gov/sites/default/files/2020-04/CPG201Final20180525.pdf

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Plans and Planning Tools

There are four (4) Plans and Planning integration tools in the toolkit, designed to address the types of plans needed at the airport and their utility across the contingency spectrum. These tools help small airports integrate contingency readiness at the planning stage:

1. Checklist of Contingency Plans

This tool lists all the types of airport contingency tools promulgated as government requirements, found in the literature, or mentioned in ACRP 03-68 interviews. With the exception of the few plans required by federal regulations for a given type of airport, the list is not prescriptive. It can serve as the basis for an airport to inventory its existing plans, identify possible gaps, and determine which plans, if any, need to be integrated using the other eight tools provided in this guide document. There is space left for an airport to add locally developed plans, plans from mutual aid partners or surrounding jurisdictions, and plans that are released after publication of this guide document.

2. Integration Decision Tree

The decision tree is intended to be used in conjunction with the Checklist of Contingency Plans. It will guide an airport to decide on which plans need to be integrated at an operational level. This potentially includes both airport plans and the contingency plans of mutual aid partners and surrounding jurisdictions. The graphical diagram captures an efficient and logical order for making these decisions.

3. Contingency Plan Deconfliction Tool

This tool enables an airport to identify commonalities among its various plans and to expose potential conflicts between plans, thereby allowing deconfliction of plans. Information captured by using this tool will also be useful when using the *Checklist & Calendar: Triggers for Plan Updates and Revisions* tool to make sure that all related or interacting parts of tools are reviewed and updated at the same time.

4. Succession Planning Tool

The succession tool serves to capture the duties, knowledges, skills, certifications, and relationships needed to perform a function should a key employee be lost through job change, retirement, or other cause. This tool could be useful at an airport of any size or type, but it will be particularly important at an airport with a small staff where everyone performs many essential functions.

Implementation Tools

There are four (4) tools designed to meet small airports' needs around integration during the "action- oriented" response and recovery phases of contingency events. These tools address how integration can be optimally executed during response behavior and identify best practices for

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updating plans. They focus on how contingency response integration optimally occurs during incident management or when airports prepare for incident management:

5. FEMA Course Selection Guide

This guide indicates which FEMA emergency training courses will be most appropriate and helpful for key personnel at an airport by job title or supervisory level. The tool can also guide the airport as to which courses to recommend to airline employees, tenants, and local government partners in order to promote a shared knowledge base and to facilitate joint training and exercising.

6. Checklist & Calendar: Triggers for Plan Updates and Revisions

This tool will help an airport ensure that its contingency plans stay current as well as optimized for action. It provides a framework for building a schedule for periodic review of all contingency plans plus identifying triggers for review and revision outside that schedule. Examples of triggers might be an incident at the airport or another airport that reveals an issue with one or more of the airport's plans. This tool should be used in conjunction with the deconfliction tool) to make sure that all related and interacting plan components are reviewed in concert and updated if necessary.

7. Checklist of Exercise Types

This checklist is a concise summary of the seven exercise types recognized by the Homeland Security Exercise and Evaluation Program (HSEEP) including the basic features, advantages, and disadvantages of each type of exercise. The tool emphasizes the two most pertinent exercise types related to contingencies at small airports: tabletop exercises (TTX) and drills. While any of the seven HSEEP exercise categories can promote plan integration and testing, TTXs and drills are the most efficient, most accessible, and easiest to turn into plan improvements.

8. EOC Tool For Integrating Airport Contingency Plans

The EOC tool will help airports consider implementing a hard copy or virtual version of the Emergency Operations Center (EOC) scaled to their size, staff, and resources. The EOC is the real-time congregation and interaction of the numerous incident response stakeholder leaders, departments and agencies during the contingency that can essentially define effective integration, but it can take many forms. An EOC can be a dedicated turn-key space on an airport, a conference room with adequate communications and display capabilities, a neighboring jurisdiction's EOC off the airport, or a virtual EOC based on a pre-established and exercised Teams, Zoom, or Meet (or other conference platform) dedicated meeting.

This tool can help an airport:

- Answer the question, "How can we best manage incident information, coordination, and action plans during the event itself?"
- Recognize that "the EOC approach" is the best model for integrated incident management by involving the right mix of decision makers by emphasizing the importance of relationships and joint practice.

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- Test whether all necessary EOC functions are being addressed.
- Present the possible configurations of effective EOCs—i.e., the right mix of "seats" (virtual, ad hoc, or physical) are occupied with airport, city/county and other stakeholder organizations.
- Address situational data gathering and collaborative decision-making.

Communication Tool

There is one (1) Communication tool designed to help airport management ensure that communications are built into each of their plans to meet regulatory, legal, operational, and public interest needs.

9. Crisis Communications Insert Template for Contingency Plans

This template provides a format and process for inserting an appropriate message for the public, the media, or internal use when something happens in an incident that requires communication. The tool is designed to be completed quickly and inserted into whichever airport contingency plan is guiding the current stage of the response or recovery effort. The tool serves as a reminder of the parties involved, the nature of the audience for the messaging, and means or locations where the message can be communicated.

Summary

These tools were crafted to be straightforward, helpful, and easy-to-use. Wherever possible we developed checklists or visuals aids rather than narrative-based tools. The tools can be downloaded from the Transportation Research Board's website in printable forms so they accessible in the event of a power outage. In the most extreme circumstances, it is possible that the entire response team at a small airport could be staffed by individuals with no prior experience or training in response and recovery. These tools were designed to guide them if necessary.

The tool kit addresses the needs of small airports that research from ACRP 03-68 revealed. It should help small airports identify the gaps in their resiliency planning and improve how they think about integrating their contingency plans internally and with city/county plan.

While some small airports may need to use all nine tools, the toolset is provided as a suite of individual tools that can become a resource from which airports can choose those that can specifically help them improve how they choose to integrate contingency planning.

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Tools

CHECKLIST OF CONTINGENCY PLANS

This checklist provides a wide range of contingency plans that small airport leaders might consider for development and implementation.[1] It is designed to help airports answer the question "*What contingency tools do we currently have in place?"* and to note if they are required, document if they are stand-alone or a component of a larger plan and note the "owner" of each.

Note – The airport may want to consider using a Threat and Hazard Identification and Risk Assessment (THIRA) to help identify and prioritize the plans you'll be considering below. THIRAs are a three-step risk assessment process that helps the airport understand your risks and what you need to do to address those risks by addressing:

- What threats and hazards can affect the airport?
- What impacts would those threats and hazards have on the airport if they occurred?
- Based on those impacts, what capabilities should we have in place and what plans should we prioritize?[2]

Instructions:

1. Indicate the plans you have in place with a Yes/No (Y/N) in column 1.
2. Check the appropriate columns to characterize each plan.
3. Use the results to consider:
    a. other contingency plans that might be useful at the airport, and
    b. if there might be value in any of these plans being integrated.

Checklist of Contingency Plans

Need Y/N	Contingency Plan	Regulatory Requirement	Part of City/County Plan	Stand-Alone Airport Plan	Annex	Responsible Staff Title or Name
Administrative						
	Airport Strategic Plan					
	Airport Master Plan					
	Hazard Mitigation Plan					
	Contingency Review Plan					
	Annual Work Plan					
	Cooperative Plans					
	Crisis Management Plan					
	Exercise Plan					
	Training Plan					
	Resiliency Plan					
	Sustainability Plan					

[1] These were identified during ACRP 03-68 case study interviews by members of the small airport community.
[2] https://www.fema.gov/sites/default/files/2020-04/CPG201Final20180525.pdf

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Checklist of Contingency Plans						
Need Y/N	Contingency Plan	Regulatory Requirement	Part of City/County Plan	Stand-Alone Airport Plan	Annex	Responsible Staff Title or Name
Emergency Response						
	Airport Emergency Plan (AEP)					
	Terminal Evacuation Plan					
	Active Shooter Plan					
	Water Rescue Plan					
	Emergency Response Plan (ALL)					
	Crisis Communications Plan					
	Terminal Evacuation Plan					
Response						
	SOP – Hazard Specific					
	• Aircraft Accident					
	• Bomb Incidents/Threat					
	• Structural Fires					
	• Fuel/Fuel Farm Fires					
	• Natural Disasters					
	• Hazardous Materials					
	• Sabotage, Terrorism					
	• Power Failure					
	• Water Rescues					
	• Evacuation					
	Field Operations Guides					
	Tactical Worksheets					
	Standard Operating Procedures (SOP), Field Operations Guides (FOG), Tactical Worksheets[3]					
	Prepositioning Materials and Equipment for Response and Recovery					
Recovery						
	Business Continuity Plan					
	Continuity of Operations Plan					
	IT Disaster Recovery Plan					
	Family Assistance					
	Reunification and Support					

[3] These are essentially synonymous, typically hazard-specific plans at the role level.

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Checklist of Contingency Plans						
Need Y/N	Contingency Plan	Regulatory Requirement	Part of City/County Plan	Stand-Alone Airport Plan	Annex	Responsible Staff Title or Name
	Precontracting for Goods and Services for Recovery[4]					
	Terminal Repopulation Plan					
	Recovery Plan					
Mutual Aid						
	Mutual Aid Agreements					
	Letters of Agreement					
	Memoranda of Understanding					
	Mutual Aid Plans					
	Emergency Operations Plan (regional)					
	Hazard Mitigation Plan (regional)					
Security						
	Airport Security Plan[5]					
	(Un)Planned Assemblies					
	Special Events, VIP Visit					
Technology/Systems						
	IT and Cyber Plans					
	Cyber Incident Response Plan					
Operational						
	Irregular Operations Plan (IROPS)					
	Pandemic Plan					
	Communicable Disease Plan					
	COVID Plan					
	Snow Removal Plan					
	Hurricane Plan					
	Parking Lot Overflow Plan					
	Drone Plan FAA					
	Drone Plan TSA					
	HAZMAT Remediation Plan					

[4] Technically not a plan, but a program or preparatory action that may interact with other plans.
[5] These plans are included in the table as part of the universe of airport plans; however, are not well suited for integration with other plans because they likely contain sensitive security information or SSI.

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Checklist of Contingency Plans						
Need Y/N	Contingency Plan	Regulatory Requirement	Part of City/County Plan	Stand-Alone Airport Plan	Annex	Responsible Staff Title or Name
	Housing Plans during disruptions (weather, e.g.) and stranded passengers					
Annexes and Functional Annexes						
	Functional annexes to AEP					
	Wildland Fire					

Other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans are:

- ACRP Report 16 *Guidebook for Managing Small Airports - Second Edition.*
- ACRP Report 20 *Strategic Planning in the Airport Industry.*
- ACRP Report 22 *Helping Airport and Air Carrier Employees Cope with Traumatic Events.*
- ACRP Report 32 *Guidebook for Addressing Aircraft/Wildlife Hazards at General Aviation Airports.*
- ACRP Report 65 *Guidebook for Airport Irregular Operations (IROPS) Contingency Planning.*
- ACRP Report 91 *Infectious Disease Mitigation in Airports and on Aircraft.*
- ACRP Report 93 *Operational and Business Continuity Planning for Prolonged Airport Disruptions.*
- ACRP Report 106 *Being Prepared for IROPS: A Business-Planning and Decision-Making Approach.*
- ACRP Report 112 *Airport Terminal Incident Response Planning.*
- ACRP Report 160 *Addressing Significant Weather Impacts on Airports: Quick Start Guide and Toolkit.*
- ACRP Report 171 *Establishing a Coordinated Local Family Assistance Program for Airports.*
- ACRP Report 189 *Design Considerations for Airport EOCs.*
- ACRP Report 209 *Integrating Sustainability Planning and the Environmental Review Process.*
- ACRP Synthesis 45 *Model Mutual Aid Agreements for Airports.*
- ACRP Synthesis 60 *Airport Emergency Post-Event Recovery Practices.*
- ACRP Synthesis 72 *Tabletop and Full-Scale Emergency Exercises for General Aviation, Non-Hub, and Small Hub Airports.*ACRP Synthesis 73 *Emergency Communications Planning for Airports.*
- ACRP Synthesis 82 *Uses of Social Media in Operational Response and Recovery during Airport Emergencies.*
- ACRP Synthesis 83 *Preparing Airports for Communicable Diseases on Arriving Flights.*

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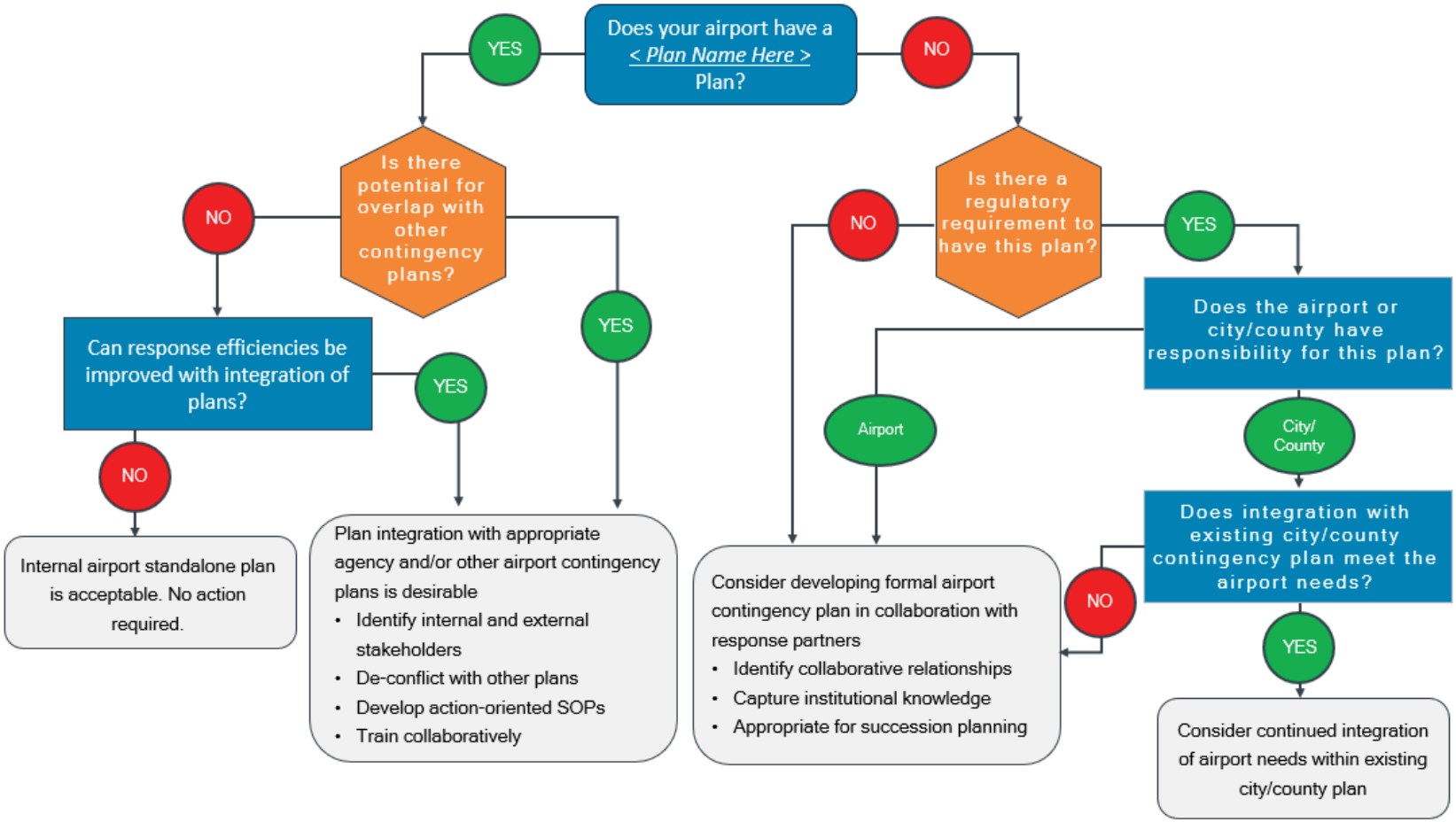
AIRPORT CONTINGENCY PLAN DEVELOPMENT AND INTEGRATION DECISION TREE

This decision tree guides small airports in considering what kinds of contingency plans they already have in place and whether they should consider developing additional contingency plans or integrating their plans. It is designed to help airports answer the questions "*What other tools do we need*?" *Should they be integrated so they don't conflict*?"

The decision tree forces binary answers to basic questions to help make decisions about plans, plan responsibility and plan integration.

Instructions:

1. Insert the name of any plan you want to evaluate, in the top blue box.
2. Follow the binary logic down the paths indicated by your answers.
3. Consider the guidance provided in the gray boxes, in deciding how to proceed with new plans and integrating your plans and those of your municipal and county partners.



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CONTINGENCY PLAN DECONFLICTION TOOL

This template is a planning tool that enables users to input any airport contingency plan (Plan A) and then assess plan commonalities, conflicts and identify resource needs and improvements against any other internal or external plan used by the airport (Plans B,C, etc.).

It is designed to answer the question "*Are all the contingency functions that are important to the airport present in our plans*?" through a disciplined way to discern the functional commonality among plans. The tool addresses common resources shared by plans and helps identify what other resources might airports need. It also helps users document elements essential for plan integration.

When deciding whether to integrate the various contingency plans at the airport, it will be helpful to determine what the plans have in common, if they conflict, or if additional resources may be needed to improve overall efficiency. This tool assists with this analysis and identification process.

Instructions:

1. Determine the airport plan to assess, inserting the plan name in the Plan A cell.
2. Then list the other plan(s) (Plans B, C, D…) that you want to compare to Plan A.
3. Next, identify what resources and procedures the plans have in common.
4. Identify possible conflicts to consider between/among the plans, using the functional considerations listed on the following page.
5. Finally, identify possible resources or gaps needed to address overall improvement, and use this as your guide to de-conflicting your plans, cross-referencing them, and addressing resource implications.

TEMPLATE TO ASSESS COMMON RESOURCES, CONFLICTS & IMPROVEMENTS AMONG PLANS				
Airport Contingency Plan		**Commonality Aspects and Common Resources**	**Plan Conflicts**	**Resource Needs and Improvements**
Plan A				
Comparable Internal/External Plan(s)				
Plan B				
Plan C				
Plan D				

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Functional Considerations for Analysis

Example Functional Areas

- Command and Control
- Communication
- Alert Notification and Warning
- Emergency Public Information
- Protective Actions
- Law Enforcement
- Firefighting and Rescue
- Health and Medical Services
- Resource Management
- Airport Operations and Maintenance – Equipment & Personnel
- Emergency Management
- Disaster Recovery
- Family Assistance Facility
- Media Facility
- Facilities Management
- Mass Absentee Planning
- Supply Chain Management
- Health and Safety
- Knowledge Management
- Security
- Crisis Communication
- Identification of Essential Functions
- Identification of Continuity Personnel
- Vital Records Management
- Continuity Facilities
- Continuity Communications
- Leadership and Staff (Succession, Delegation of Authority, Human Capital)
- Test, Training, and Exercise Program

Example Cybersecurity Plan Functions

- Operations
- Access Control
- Planning and Prevention
- Project Support
- Manage Communications to Stakeholders
- Mitigate Exposure/Risk Management
- Employee Training
- Timely Recovery

Example Pandemic Plan Functions

- Preparedness Framework
- Monitoring
- Financial Implications
- Cleanliness Protocols
- Impact to HR
- Communications
- Impact on business operations
- Preparedness Exercise
- Impact of Regulatory Requirements

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SUCCESSION PLANNING TOOL

This is a tool to assist small airports with documenting the knowledge and competencies of key personnel in their roles and responsibilities for operational level contingency planning and response. It is designed to help airports capture this information before an employee leaves the organization, so other employees have some documentation to help them close the knowledge gap when a key employee with contingency planning responsibilities leaves.

(Documenting the knowledge and practices of key airport employees is always a good succession planning element of the airport's continuity of operations plan [COOP] or business continuity plan [BCP]. This tool is specifically designed to document key airport employees' knowledge of contingency planning and response.)

Because succession planning is crucial at small/general aviation (GA) airports, it is important that institutional knowledge should not be allowed to be lost with staff retirements, resignations, family care or parental leave absences. This succession plan tool allows experienced employees to complete a form that identifies categories of knowledge they each possess. It helps airports answer the question "*How do we know how to efficiently and seamlessly respond to an airport event if a key employee were to abruptly leave their position and take their knowledge with them?*"

This tool is meant to be customizable depending on each airport's unique set of circumstances and needs.

Instructions:

1. Fill in the form as illustrated, based on individual airport needs.
2. Consider completing form collaboratively with each key employee to identify answers to open ended questions and input fields. This may help senior administrative staff to identify needs not formalized in airport owned or integrated contingency plans.
3. Develop a schedule to routinely update file form, so you keep it current.
4. Consider organizing each file form by employee position or by contingency plan.

KEY STAFF TURNOVER FILE FORM

Airport Contingency Plans

Airport: _____

Date: _____

POSITION IDENTIFICATION

Position: _____

Position Currently Held By (Name): _____

Position Reports to: _____

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Certifications/Licenses (required to perform any tasks):

_____
_____
_____
_____

CORE COMPETENCIES (check all that apply)

- ☐ Planning and Administration
- ☐ Airport Operations
- ☐ Project Management
- ☐ Emergency Response
- ☐ Health and Medical
- ☐ Supervisory Experience
- ☐ Level of ICS Training/FEMA Course Required of this Position (check all that apply):
  - ☐ ICS-100
  - ☐ ICS-200
  - ☐ ICS-300
  - ☐ ICS-400
  - ☐ ICS-700
  - ☐ ICS-800
  - ☐ G-402
- ☐ _____
- ☐ _____

TECHNICAL COMPETENCIES (check all that apply)

- ☐ Incident Command
- ☐ Public Information Officer
- ☐ Section Team Lead
- ☐ All-Hazard Response
- ☐ ARFF
- ☐ EMT/Paramedic
- ☐ Information Technology
- ☐ Heavy Equipment Operator
- ☐ Pilot Ratings
- ☐ Access Control
- ☐ _____
- ☐ _____
- ☐ _____
- ☐ _____

IDENTIFY SUBSTITUTE OR REPLACEMENT PERSONNEL (by order of succession using name or position)

Primary Replacement _____
Secondary Replacement _____
Tertiary Replacement _____

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IDENTIFY CONTINGENCY PLANS THAT EMPLOYEE PLAYS A ROLE IN (check all that apply)

☐ Airport Emergency Plan (AEP)	Physical Location: _____ Digital Location: _____
☐ Cyber Incident Response Plan	Physical Location: _____ Digital Location: _____
☐ Terminal Evacuation Plan	Physical Location: _____ Digital Location: _____
☐ Pandemic Plan	Physical Location: _____ Digital Location: _____
☐ _____	Physical Location: _____ Digital Location: _____
☐ _____	Physical Location: _____ Digital Location: _____

KEY ROLES AND RESPONSIBILITIES BY PLAN (form includes two plans, insert as many as needed)

AIRPORT CONTINGENCY PLAN #1: _____

Employee Roles and Responsibilities:

_____
_____
_____

Plan Activation/Deactivation Procedure Notes:

_____
_____
_____

How are Essential Plan Functions Performed:

_____
_____
_____

Plan Integration Notes

_____
_____
_____

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Employee's "Go-To" Contingency Contacts

ESSENTIAL OPERATIONAL, RESPONSE, AND RECOVERY CONTACT LIST			
Contact Name	Organization	Email Address	Phone Number

Supplies & Equipment the Employee Controls, Procures and Uses in Their Contingency Role

SUPPLIES & EQUIPMENT	
Supply/Tools	Location

Other Resources the Employee Uses in Their Contingency Planning and Response Role

RESOURCES			
Resource	Organization	Contact Name	Phone Number

Notes:

_____
_____
_____

EMPLOYEE COMMENTS & NOTES RELEVANT TO CONTIGENCY PLANNING AND RESPONSE ROLE

_____
_____
_____
_____

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_____
_____
_____

RECORD OF REVSION AND REVIEW		
DATE	SECTION REVISED	INITIALS

AIRPORT CONTINGENCY PLAN #2: _____

Employee Roles and Responsibilities:

_____
_____
_____

Plan Activation/Deactivation Procedure Notes:

_____
_____
_____

How are Essential Plan Functions Performed:

_____
_____
_____

Plan Integration Notes

_____
_____
_____

Employee's "Go-To" Contingency Contacts

ESSENTIAL OPERATIONAL, RESPONSE, AND RECOVERY CONTACT LIST			
Contact Name	Organization	Email Address	Phone Number

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Supplies & Equipment the Employee Controls, Procures and Uses in Their Contingency Role

SUPPLIES & EQUIPMENT	
Supply/Tools	Location

Other Resources the Employee Uses in Their Contingency Planning and Response Role

RESOURCES			
Resource	Organization	Contact Name	Phone Number

Notes:

_____
_____
_____

EMPLOYEE COMMENTS & NOTES RELEVANT TO CONTIGENCY PLANNING AND RESPONSE ROLE

_____
_____
_____
_____
_____
_____
_____

RECORD OF REVSION AND REVIEW		
DATE	SECTION REVISED	INITIALS

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Other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans are:

- ACRP Report 186 *Guidebook on Building Airport Workforce Capacity*
- ACRP Report 20 *Strategic Planning in the Airport Industry*
- ACRP Synthesis 11-03/Topic 06-07 (underway) *Impacts of COVID to Airport Work Models and Strategies*
- ACRP Synthesis 18 *Aviation Workforce Development Practices*
- ACRP Web-Only Document 28 *Identifying and Evaluating Airport Workforce Requirements*
- ACRP Report 93 *Operational and Business Continuity Planning for Prolonged Airport Disruptions*

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FEMA COURSE SELECTION GUIDE

This guide helps small airports determine how best to train their multi-tasking personnel in the Incident Command System (ICS) as a widely used structure to assign roles and responsibilities during contingency responses. It is designed to answer the question "*What is the right training approach to incident management, so our staff understands basic incident management and action plans*?"

Airline personnel and other stakeholders are included in this guide to prompt airport leaders to think about the value of this training to those working together in the airport environment. At a small airport, the likeliest situation will have mutual aid partners or a city/county working with the airport to manage part or all of some functions. At airports with a small staff, roles may be combined or may supplemented by off-airport expertise. This puts a premium on a shared knowledge base such as provided by these FEMA courses and through joint training and exercises.

Instructions:

1. Locate the description of the roles/position that best matches your personnel.
2. Review the summary descriptions on the following pages of each course indicated for each role to confirm that this is appropriate course work and knowledge for each participant's expect role during contingency response.
3. Go to https://training.fema.gov/nims/ for more information.

FEMA Course Number	ICS-100	ICS -200	ICS-300	ICS-400	IS-700	IS-800	G-402
ROLE/POSITION							
Emergency Manager (full-time or collateral duty)	**X**	**X**	**X**	**X**	**X**	**X**	
First Line Supervisors (Operations Supervisors, Security Coordinators, Law Enforcement Captains and Lieutenants)	**X**	**X**			**X**		
Mid-level Management (Director of Customer Service/Experience, Airline Station Managers, TSA Liaison/Station Manager, Managers of Tenants)	**X**	**X[1]**	**X[1]**		**X**		

[1] Depending on possible involvement in incident.

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FEMA Course Number	ICS-100	ICS -200	ICS-300	ICS-400	IS-700	IS-800	G-402
Command Staff (Director of Operations, ARFF Chief, Police Chief, Director of Security, Director of Facilities, Director of Maintenance, Public Information Officer)	**X**	**X**	**X**	**X**	**X**	**X**	
Policy Makers (CEO/Executive Director/Aviation Director, COO, CFO, CIO, Board Members; possibly also HR Director, General Counsel, Airport Development Director, Risk Advisor)	**X**				**X**		**X**

CLASS DESCRIPTIONS

ICS-100: Introduction to the Incident Command System, introduces the Incident Command System (ICS) and provides the foundation for higher level ICS training. This course describes the history, features and principles, and organizational structure of the Incident Command System. It also explains the relationship between ICS and the National Incident Management System (NIMS).

ICS-200: Basic Incident Command System for Initial Response, reviews the Incident Command System (ICS), provides the context for ICS within initial response, and supports higher level ICS training. This course provides training on, and resources for, personnel who are likely to assume a supervisory position within ICS.

ICS-300: Intermediate ICS for Expanding Incidents, provides an in-depth focus on the NIMS Incident Command System (ICS) that includes the tools, practices, and procedures that are available in ICS to effectively manage emergency incidents or planned local events. Designed for those emergency response personnel who would function in a Command or General Staff position during an incident or event.

ICS-400: Advanced ICS Command and General Staff, emphasizes large-scale organizational development, roles and relationships of the command and general staff, and planning, operational, logistical, and fiscal considerations related to large and complex incident and event management.

ICS-700: An Introduction to the National Incident Management System, defines the comprehensive approach guiding the whole community - all levels of government, nongovernmental organizations (NGO), and the private sector - to work together seamlessly to prevent, protect against, mitigate, respond to, and recover from the effects of incidents. The course provides learners with a basic understanding of NIMS concepts, principles, and components.

G-402: NIMS Overview for Senior Officials, familiarizes Senior Officials (executives, elected and appointed officials) with their role in supporting incident management within the NIMS framework.

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CHECKLIST AND CALENDAR: TRIGGERS FOR PLAN UPDATES AND REVISIONS

This tool helps airports schedule their contingency plans for review and revision, so they remain current and optimally actionable.[1] It is designed to help airports answer the question "*How often should we change our plans*?" It is a best practice to review any plan activated during an actual incident as soon as possible after the incident. This is most often accomplished as part of the after-action review / improvement plan (AAR/IP) process. This review should be in addition to the normally scheduled review for that plan, but an airport should use its judgment on the allocation of staff time and effort to reviewing plans that have been recently activated and reviewed.

The list of contingency plans comes from the ***Checklist of Contingency Plans Tool*** with the addition of the Airport Security Program (ASP) and Airport Financial Contingency Plan. The list of recommended exercises to test revisions comes from the ***Checklist of Exercise Types Tool***. An airport using this scheduling tool can add or delete any non-required plans to reflect its unique circumstances.

Most small airports will have 12 or fewer contingency plans, so an efficient practice would be to revise one per month. One possible schedule would be to take the month of the required emergency plan review and exercise (tabletop or full-scale) as "month zero" and then update one plan per month for the rest of the year.

One person should be tasked with tracking the completion of the reviews and, when indicated, revisions. If the airport has an emergency manager, this would be an appropriate assignment. If there is no emergency manager, the ARFF chief or operations director may be the best person. At very small airports, the airport manager or assistant manager would need to track this process.

Instructions:

1. For each plan, there is a line item in which to enter frequency of review, planned date of review, and the lead responsibility for review.
2. It also suggests drill and exercise types for airports to consider for testing revised plans.
3. The tool also indicates which plan reviews and updates are required by regulation.

CHECKLIST OR CALENDAR FOR BEST PRACTICES FOR PLAN UPDATES AND REVISIONS					
Contingency Plan	**Review and Revision Required**	**Freq. & Month for Review**	**Responsible Party**	**Recommended Exercise(s) to Test Revision**	**Other Triggers for Plan Updates**
Airport Emergency Plan (AEP)	Yes (for Part 139 airports)	Annual	Varies	Frequent Seminar and mini-TTX leading up to main TTX (years	When any one of these plans is updated, refer to the Tool #3 (Plan Deconfliction Tool)

[1] A tip of the hat to Tallahassee International Airport (TLH) who showed the research team this tool.

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CHECKLIST OR CALENDAR FOR BEST PRACTICES FOR PLAN UPDATES AND REVISIONS					
				1 and 2) and full-scale triennial (year 3)	to identify all other plans that should be reviewed and possibly updated
Airport Security Program Contingency Plan	Yes (for Class II and larger airports	Annual	ASC	Frequent Seminar and mini-TTX leading up to main annual TTX	
Airport Financial Contingency Plan					
Business Continuity Plan (BCP)					
Continuity of Operations Plan (COOP)					
IT Disaster Recovery Plan					
Irregular Operations Plan (IROPS)	Yes (for Part 139 airports)				
Mutual aid plans					
Cooperative Plans					
Cyber Incident Response Plan					
Crisis Management Plan					
Crisis Communications Plan					
Recovery Plan					
Precontracting for goods and services for recovery					
Terminal Evacuation plan					
Terminal Repopulation Plan					

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CHECKLIST OR CALENDAR FOR BEST PRACTICES FOR PLAN UPDATES AND REVISIONS					
Communicable Disease Plan					
Pandemic Plan					
COVID plan					
Housing plans during disruptions (weather, etc.) stranded passengers					
HAZMAT remediation					
Drone Plan/UAS Plan (FAA)	Yes (AEP)				
Drone Plan/UAS Plan (TSA)	Yes (ASP)				
Family assistance/reunification plan					
Active shooter/active assailant plan	Yes (AEP)				
Parking Lot Overflow Plan					
Emergency Operations Plan (regional)	Varies with state				
Hazard Mitigation Plan (regional)	Varies with state				
Annexes and Functional Annexes [to Emergency Management Plan]	Yes (as part of AEP)	Annual		Frequent seminar and mini-TTX leading up to main TTX (years 1 and 2) and full-scale triennial (year 3)	
Hazard-Specific Tactical Worksheets/Standard Operating Procedures (SOPs)/Field Operating Guides (FOGs)					

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CHECKLIST OR CALENDAR FOR BEST PRACTICES FOR PLAN UPDATES AND REVISIONS					
Resilience/Resilience Plan					
Sustainability Plan					
Airport Master Plan (cross-reference contingency plans)	Yes	5 years			
Airport Strategic Plan (cross-reference contingency plans)					
Contingency Plan Review Plan / Annual Work Plan					
Exercise Plan					
Training Plan					
Other Local Plan					
Other Local Plan					
Other Local Plan					
Other Local Plan					

Other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans are:

- ACRP Report 12 *An Airport Guide for Regional Emergency* **Planning** *for CBRNE Event.*
- ACRP Report 20 *Strategic Planning in the Airport Industry.*
- ACRP Report 65 *Guidebook for Airport Irregular Operations (IROPS) Contingency Planning.*
- ACRP Report 93 *Operational and Business Continuity Planning for Prolonged Airport Disruptions.*
- ACRP Report 106 *Being Prepared for IROPS: A Business-Planning and Decision-Making Approach.*
- ACRP Report 112 *Airport Terminal Incident Response Planning.*
- ACRP Report 113 *Guidebook for General Aviation Facility Planning.*
- ACRP Report 147 *Climate Change Adaptation Planning: Risk Assessment for Airports.*
- ACRP Report 188 *Using Existing Airport Management Systems to Manage Climate Risk.*
- ACRP Report 209 *Integrating Sustainability Planning and the Environmental Review Process.*
- ACRP Synthesis 73 *Emergency Communications Planning for Airports.*

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CHECKLIST OF EXERCISE TYPES

This guide helps airport leaders decide what type of exercises are optimal at various stages of contingency planning. It is designed to answer the question "*How often should we conduct exercises and for what purposes*?" Exercises are an excellent way for airports to improve integration of contingency plans by inviting the participation of planners and responders from city, county and other stakeholder organizations.

FEMA's Homeland Security Exercise and Evaluation Program (HSEEP) provides a set of guiding principles and an approach for planning, conducting, and evaluating exercises. This guide provides information to help airports select the best exercise for their purposes.

There are two categories of exercises:

- **Discussion-based Exercises** include seminars, workshops, tabletop exercises, and games. These types of exercises familiarize players with or develop new plans, policies, procedures, and agreements.
- **Operations-based Exercises** include drills, functional exercises, and full-scale exercises. These exercises validate plans, policies, procedures, and agreements; clarify roles and responsibilities; and identify resource gaps. Operations-based exercises include a real-time response such as initiating communications or mobilizing personnel and resources.

NOTE: Of the seven types of exercises, the two most useful for integrating contingency plans are tabletop exercises (TTX) and drills. The descriptions of these two types are highlighted in yellow below. In general, the most productive approach is to develop a TTX then select pieces (core capabilities, objectives, injects, or other specific things you wish to test) and turn them into a drill.

Instructions:

1. Review each of the four types of exercises.
2. Consider both the benefits and considerations of each.
3. Select the optimal exercise for each use case at the aiport.
4. Download HSEEP templates and instructions at https://preptoolkit.fema.gov/web/hseep-resources to help you design and facilitate the exercises you select.

Seminar

- A discussion-based exercise that orients participants to or provides an overview of plans, policies, procedures, protocols, and resources.
- Lecture-based, usually led by a seminar facilitator/presenter with limited feedback or participation from attendees.
- A good starting point for developing or making major changes to existing plans.

Benefits: Minimal time constraints; effective for small or large groups.

Considerations: Less engaging for participants.

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Workshop

- A discussion-based exercise often employed to develop policies, plans or procedures.
- A group based discussion, often facilitated by subject matter expert who guides discussion.
- Clearly defined objectives and goals, focusing on a specific issue, such as the development or revision of a specific plan.

Benefits: Effective for small or large groups; more participant engagement.

Considerations: Specific objectives need to be identified in advance.

Tabletop

- A discussion-based exercise in response to a scenario, intended to generate a dialogue of various issues to identify strengths and areas for improvement, and/or achieve changes in perceptions about plans, policies, or procedures.
- Scenario is presented to describe an event at a simulated time.
- Players apply their knowledge and skills to a list of problems presented by the facilitator.
- Problems are discussed as a group, and resolution may be reached and documented for later analysis.

Benefits: Enhance understanding of roles and responsibilities during a defined incident; validate plans and procedures.

Considerations: Requires more advanced planning and an experienced facilitator; need to create a problem-solving environment.

Game

- A discussion-based exercise that is a structured form of play guided by clear rules, data, and procedures for its execution.
- Games are designed to depict an actual or hypothetical situation to ensure that the participants make decisions and take actions that would be plausible.
- Games can be used to reinforce training, stimulate team building, or enhance operational and tactical capabilities.

Benefits: Open, decision-based format of a game can incorporate “what if” questions; no actual resources are used.

Considerations: Requires extensive planning and pre-scripted activities.

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Drill

- An operations-based exercise often employed to validate a specific function or capability in a single agency/organization.
- Used to practice and maintain skills, evaluate new procedures or policies, and determine if plans can be executed as designed.
- Results are measured against established standards.

Benefits: Participants receive immediate feedback. The narrow focus requires minimal pre-planning.

Considerations: Clearly defined plans, procedures, and protocols need to be in place

Functional Exercise

- An operations-based exercise designed to test and evaluate capabilities and functions while in a realistic, real-time environment; however, movement of resources is usually simulated.
- Events are projected through a realistic exercise scenario with event updates that drive activity typically at the management level.
- Controllers typically use a Master Scenario Events List (MSEL) to ensure participant activity remains within predefined boundaries.
- Evaluators observe behaviors and compare them against established plans, policies, procedures, and standard practices.

Benefits: Inter-department and inter-agency roles and responsibilities are validated and real-world relationships are strengthened; EOC and Command Post are evaluated; engaging for participants.

Considerations: Deployment of resources and personnel usually simulated but demands on exercise personnel is high. This exercise requires much pre-planning and coordination amongst planners.

Full-Scale Exercise

- An operations-based exercise that is typically the most complex and resource-intensive of the exercise types and often involves multiple agencies, jurisdictions/organizations, and real-time movement of resources.
- Events are projected through an exercise scenario with event updates that drive activity at the operational level.
- Involves multiple agencies, organizations, and jurisdictions and use of MSEL drives player actions.
- Often includes many players operating under cooperative systems such as the Incident Command System (ICS) Unified Command.

Benefits: Conducted in a realistic environment to mirror a real incident by presenting complex problems requiring multi-agency, multi-department response.

Considerations: Exercise site is usually large with many activities occurring simultaneously; mobilization of personnel and resources is required. Extensive planning and coordination among parties is required.

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Other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans are:

- ACRP Synthesis 72 *Tabletop and Full-Scale Emergency Exercises for General Aviation, Non-Hub, and Small Hub Airports.*

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EOC TOOL FOR INTEGRATING AIRPORT CONTINGENCY PLANS

The Emergency Operations Center (EOC) is the real-time congregation and interaction among the numerous incident response stakeholder leaders, departments and agencies during a contingency that can essentially define effective integration, and it can take many forms. This operational tool will help airports consider implementing a hard copy or virtual version of the Emergency Operations Center (EOC) scaled to their size, staff, and resources.

This EOC educational aid answers the question, "*How can we best manage incident information, coordination, and action plans during the event itself*?" The tool also:

- Recognizes that "the EOC approach" is the best model for integrated incident management by involving the right mix of decision makers.
- Tests whether all necessary EOC functions are being addressed.
- Presents the possible configurations of effective EOCs—i.e., the right mix of "seats" (virtual, ad hoc, or physical) are occupied with airport, city/county and other stakeholder organizations.
- Addresses situational data gathering and collaborative decision-making.

Best Practices for Managing Incident Information, Coordination, and Actions in Response and Recovery

The established doctrine for the best practice of managing and coordinating actions and information flows during incidents ideally involves three elements - an EOC, one or more Command Posts, and a Senior Policy Group.

In reality, the size of the incident and the resources available to an airport often dictate that the three elements collapse into a single group or even a single individual. At a small airport, each person in the EOC will wear more than one hat and perform multiple types of functions. For this reason, the most productive approach is to work from necessary EOC functions towards a structure that is possible given the airport's resources.

Instructions:

1. Review the essential functions of the EOC and supporting detail.
2. Note if each are addressed in your EOC, an EOC the airport uses with city/county responders, or informally in your contingency response protocol.
3. For this to work it is essential that each person filling these functions have both the appropriate FEMA training (see **FEMA Course Selection Guide**) and have had sufficient opportunities to practice the roles in seminar, tabletop, functional, and full-scale exercises (see **Summary of Exercise Types** tool).

What Are the Necessary Functions of the EOC?

The Incident Command System (ICS) provides clear guidance on the categories of functions that are likely to be required of an EOC during response to, and recovery from an incident.

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Necessary Functions of the EOC		
Function	**Supplemental Detail**	**Addressed?**
Command	Unified Command, Incident Commander, Policy Group	
Operations	Operations, Situational Awareness, Communications	
Planning	Planners, Development Officer, Persons knowledgeable about infrastructure of the airport	
Logistics		
Finance/Administration	Purchasing, Procurement, Documentation	
Safety		
Liaison	Mutual Aid Partners, Commercial Development Manager (who are highly familiar with all tenants)	
Public Information	Joint Information Center, crisis communications	

For example, at a non-hub primary airport with a full-time staff of 10 employees, the airport director might perform command, planning, liaison, and public information functions while the director of operations or ARFF chief might handle operations and safety, and the office manager might handle finance/administration and logistics. In this example, three managers wearing a total of eight hats handle all eight necessary functions. For this to work it is essential that each person have both the appropriate FEMA training **(see FEMA Course Selection tool)** and have had sufficient opportunities to practice the roles in seminar, tabletop, functional, or full-scale exercises (**see Summary of Exercise Types tool**).

Possible Configurations of Effective EOCs

An airport may have one or more of these locations and types of EOCs available to it. If so, the airport should try to decide in advance which arrangement will be used for each type of incident. If an airport does not have access to any of these possibilities, it should move expeditiously to develop at least one of them. In most cases, the quickest and most economical solution will be to piggy-back on a local jurisdiction's EOC. If this happens, joint training and exercising will become especially important.

NOTE: Regardless of the physical EOC an airport may have at the airport or have access to off-airport, it is highly advisable to have a pre-established and exercised Teams, Zoom, or Meet dedicated meeting to serve as the EOC. This will be useful if the physical EOC is damaged, inoperable, or inaccessible or if the travel time of essential personnel to reach the EOC is a problem.

During an incident key airport personnel will not want to leave the airport to go to an off-airport EOC. Those who are not at the forward command post will convene in an office or conference room with internet, phones, and suitable environment. This group would then interact with the off-airport EOC. As already stated, in this case joint training and exercising become essential.

Instructions:

1. Identify whether you have access to an on or off-airport location for your EOC.
2. Next, consider the Configuration components necessary for the effectiveness of each, adding as many lines as you need.
3. Finally, note of these elements are available or need to be newly planned for.

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Possible Configurations of an Effective EOC		
Location	**Specific Configuration**	**Available?**
On-Airport	Dedicated Light-Switch EOC (Always ready to activate)	
	Dedicated EOC Space	
	Multipurpose Space Pre-wired and Pre-Equipped for EOC Use	
	Multipurpose Space with Adequate Connectivity	
	Virtual EOC/Distributed in Multiple Spaces	
	Mobile Command Post	
	Trunk-top at Command Post	
	Other (Specify):	
Off-Airport	City EOC	
	County EOC	
	Regional EOC	
	State EOC	
	Federal Partner EOC	
	Mutual Aid Partner Agency EOC	
	Mobile Command Post	
	Virtual EOC (e.g., Web EOC or dedicated Teams, Zoom or Meet channel)	
	Other (Specify):	

EOC Essential Capabilities

The airport's EOC—whether owned by the airport or using another jurisdiction's EOC—must have the capabilities that are essential to support situational data gathering and collaborative decision-making.

Instructions:

1. Review the fundamental and optional EOC capabilities below.
2. Note if the individual capabilities are present.
3. Add as many optional capabilities as necessary to your unique airport circumstances.

Essential Capabilities of an EOC		
Importance	**Capability**	**Have?**
Fundamental	Situational Awareness	
	Robust Internal Communications	
	Robust External Communications	
	Secure Location against Threats, Risks, and Hazards	
	Physical or Virtual Arrangement Conducive to Collaboration	
	Back-up Power Supply	
	Adequate Space	
	Comfort in Case of Long Duration Activation	
	Self-sufficiency for Water, Food, and Rest Area	

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Nice to Have	Intuitive Controls	
	Private Spaces for Law Enforcement or Intelligence Interviews	

Use the combination of these three EOC component inventories to identify and locate a physical EOC you can use, newly outfit an acceptable EOC location, or devise virtual EOC capabilities.

EOC Essential Documents

The airport's EOC will need ready access to important documents (diagrams, schematics, maps, MOUs, contract lists, etc.) to support situational data gathering and collaborative decision-making. List those documents and their physical or digital location here for easy access during contingencies:

Document Name	Physical Location (office or file)	Digital Location (drive or PC)
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans are:

- ACRP Report 189 *Design Considerations for Airport EOCs.*

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CRISIS COMMUNICATIONS INSERT TEMPLATE FOR CONTINGENCY PLANS

This template helps guide airport leaders in managing its messaging of disclosures, stakeholder communications, and reporting that may be required after a contingency event or in anticipation of such an event. It is designed to answer the question "*How do we make sure we are communicating effectively and accurately to our constituencies throughout the incident action lifecycle*?"

If a contingency plan is being used to guide an airport's actions and the need arises to communicate either inwards or outwards, use this template as a plan insert or attachment. Not all communications tools listed may pertain at your airport or in the situation triggering use of the contingency plan.

Instructions:

1. For each type of contingency plan, complete and insert (or attach) this template at each place in the plan that requires (whether by policy, best practice, or legal/regulatory requirement) formal internal or outward communication.
2. Communications tools that your airport does not use may be deleted.
3. When coordinating communications around contingencies with municipal and county jurisdictions, be sure that your communications templates are synchronized with government spokespersons.

	Continency Plan:		
	Trigger Event or Stage:		
Preplanned Message:			
Who Vets Message Prior to Release:			
Who Approves Release of Message:			
Who Communicates the Message:			
Who Outside Airport Is Involved in This Messaging, i.e., involved in crafting the message or disseminating it?			
How is the Message Communicated:	**Communications Tool**	**For Internal or External Communication?**	**Applies to Airport? Y/N**
	Internal Communication App (e.g., Everbridge)		
	800 MHz Radio		
	Public Address System		
	Portable Bullhorns		
	Airport Employees in Person		
	Visual Paging		

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	Communications Tool	For Internal or External Communication?	Applies to Airport? Y/N
	Flight Information Display System (FIDS)/Baggage Information Display System (BIDS)		
	Wireless Emergency Alert (WEA)		
	Press Release		
	Press Conference		
	Road Barricades		
	Corridor/Concourse/Door Barricades		
	Other (Specify):		
	Social Media: Facebook		
	Social Media: Twitter		
	Social Media: Instagram		
	Social Media: TikTok		
	Social Media: SnapChat		
	Social Media-Other (Specify):		

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Other ACRP resources that may be helpful to an airport's efforts to integrate its contingency plans are:

- ACRP Report 22 *Helping Airport and Air Carrier Employees Cope with Traumatic Events.*
- ACRP Report 171 *Establishing a Coordinated Local Family Assistance Program for Airports.*
- ACRP Report 201 *Airport Emergency Communications for Persons with Disabilities and Others with Access and Functional Needs and Non-English Speakers Guidebook.*
- ACRP Synthesis 60 *Airport Emergency Post-Event Recovery Practices.*
- ACRP Synthesis 73 *Emergency Communications Planning for Airports.*
- ACRP Synthesis 82 *Uses of Social Media in Operational Response and Recovery during Airport Emergencies.*

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Case Study Report

Scope of Work

As Task 4 (A), the ACRP 03-68 research team conducted 10 case studies[1] with eight airports, a maritime critical infrastructure facility, and a community complex containing several airports. See Table 1. Airport size categories are based on final 2021 FAA enplanement data. GA airport categories are from FAA 2012 National Asset Study, Appendix B.

Originally, we had targeted seven airports and an additional critical infrastructure case – Missouri Southern State University (MSSU) - because of its integrated response to the Joplin tornado in 2011 with county, municipal and federal agencies, and stakeholders. Unfortunately, officials with institutional knowledge of that event separated from MSSU and could not be located for interviews. We added an eighth airport – Aspen Pitkin County Airport – with the approval of the Senior Program Officer. Because the case required contact with the ACRP 03-68 Panel Chair, we also received approval for this addition from the ACRP Manager. We determined that interviews with two non-airports would suffice for a representative sample.

TABLE 1. CASE STUDIES

	Entity/Organization	Size	State	Completed
1	Cotulla – La Salle County Airport (COT)[2]	GA Basic	TX	August 9
2	Eugene Airport (EUG)	Small Hub	OR	August 10
3	Lakeland Linder Regional Airport (LAL)	GA Reliever Regional	FL	August 19
4	Lincoln Airport (LNK)	Non-hub Primary	NE	August 31
5	Range Regional Airport (HIB)	Commercial Service	MN	July 21
6	Tallahassee International Airport (TLH)	Non-hub Primary	FL	August 4
7	Tucson International Airport (TUS)	Small Hub	AZ	August 1
8	Aspen Pitkin County Airport (ASE)	Non-hub Primary	CO	August 17
9	Port of Vancouver	Maritime Port	WA	August 11
10	State of Nevada Emergency Management via Washoe County Office of Emergency Management	Complex community context for airports (LAS, RNO, CXP, HND, RTS)	NV	August 16

In case study 10, the Washoe County complex community context for airport contingency planning, LAS is a large hub, RNO is a medium hub, CXP is a GA Reliever Regional airport, HND is a GA Reliever National airport, and RTS is a GA Reliever Local airport in the region.

[1] We additionally completed one (1) supplemental interview during August with Soteria Solutions to supplement our COT interview. Soteria Solutions is a for-profit emergency management consulting firm that LaSalle County (where COT is located) has been using to upgrade response plans and capabilities.
[2] The COT interview revealed integration by using an outside firm to foster joint consolidated preparedness activities.

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Methodology

Case Study Interviews

We employed a disciplined approach to our case studies to ensure accurate and fulsome research feedback from our targets:

- We completed as much of the literature review around each case study as we could, prior to the interviews, so we could refer to those artifacts.
- In most cases at least three research team members participated in each interview, two as questioners and note takers and one as primary scribe. In some case four team members participated, ensuring that we listened broadly to what our subjects had to say and could dive deep into areas of interest.
- Our scribe consolidated our notes for team approval before presenting them to the interview targets for correction or amplification.
- Some targets additionally provided us with sample contingency planning tools that work well at their facilities or that they consider best practices.

Review & Analysis

- Each team member individually reviewed the consolidated interview notes and sample tools received from our targets, and produced a summary analysis of the issues, considerations, and findings from their perspective.
- A Co-Principal Investigator consolidated these into a summary analysis tool that characterized each study element and suggested their potential implications for the integration tools that the team will ultimately recommend for development and testing.
- Each team member provided input, edits, and comments to that summary analysis. The research team debated open elements of the analysis, arriving at a final version that reflects the concurrence of the team.

Observations & Findings Summary

Context

Solving for a Negative

ACRP 03-68 is about how to integrate contingency planning for small airports. In none of our case study interviews, however, did we hear that airports consider plan integration as important to them or that the absence of plan integration was considered a problem. We delved into the types of plans airports have in place and in work, the problems they face keeping plans up to date and exercised, and the challenges they face from the universal issue they articulated around insufficient budgets and staff availability to try to reveal what kind of integration model might be beneficial. We realized that the absence of affirmation by the interviewees that contingency plan integration is seen as a problem, forces the research team to “work between the lines,” draw some conclusions that may not have been proffered by our case study airports, take some guidance from non-airport organizations that have considered the topic, and recommend tools that can improve how small airports develop, implement, and integrate contingency plans in practice.

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Flexibility

It is apparent from our interviews and tool reviews that small airports span numerous ownership and operating models – independent, city/county-operated, and authorities. Model diversity implies that airports consider their contingency plans collaboratively with a disparate mix of stakeholders. This obligates the research team to consider integration tools with the flexibility to work for airports in each model.

Financial Resilience Plans

The consideration of including airport financial contingency plans in our study and potential integration is interesting and presents the research team with some challenges. Typical contingency plans like airport emergency plans (AEP), irregular operations plans (IROPS), and continuity of operations plans (COOP) can inform the notion of financial resilience in several ways:

- The incident command system (ICS), when being used by airports, addresses financial elements of incident response by including a Finance Section Chief who is responsible for making pre-approved or emergency purchases of systems, materials or supplies, some of which could be financially material (such as a generator, for example).
- Access to federal and state grants also informs financial resilience by providing airports with funding over and above their standard budgeting process. It can be expected that, to the extent that plan integration yields indirect benefits from better coordination, this can manifest in financial advantage.
- After some types of large-scale crisis incidents, the availability of federal reimbursement funding and the process it may take airports to complete and submit documentation of recovery costs also speaks to an intersection of operating contingency planning and financial resilience.
- For some kinds of incidents considered in contingency plans, it may be known that there is likely to be a material or sustained impact on airport financial resilience, so plans such as a pandemic plan or recurring regional power outages naturally inform financial resilience because the crisis event is so impactful that it can shutter airport operations that generate material revenue due to a massive reduction in airport customer – for airlines, parking, dining, and on-site activities.

It is the research team's sense that, while contingency plan integration can inform and be an input into the airport's financial resilience planning, its financial resilience plan poorly integrates with the other plans that are "contingency-oriented."

Because we interviewed "action-focused" airport executives and not senior airport management or executives with overall financial or strategic responsibility for the airport, we felt it was important to additionally reach out to financial executives at several airports and supplement our understanding of the relationship of contingency planning and financial planning before making our conclusions.

The research team interviewed five airport senior managers (four chief financial officers and one CEO) to test our tentative finding that airport financial contingency plans are so different that

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they are inappropriate for integration with operational and emergency contingency plans. We interviewed CFOs at CLT (large hub), ROA (non-hub), TUS (small hub), and TLH (non-hub), as well as the CEO at GSP (small hub).

All five agreed with the finding that financial contingency plans have such different goals and time frames that they cannot productively be integrated into other types of contingency plans. However, this does not mean that there are no connections between the two types of plans. Financial contingency plans focus on risk and risk mitigation, which should be a major influence on operational and emergency contingency plans. In the other direction, the various types of operational and emergency contingencies can affect numerous aspects of financial contingency plans.

Three of the CFOs noted that the pandemic (a "black swan event," i.e., an exceedingly rare event with immense consequences) forced financial contingency planning and operational/emergency contingency planning to resemble each other far more than under normal circumstances. One CFO noted that IT contingency plans and cyber security plans could be integrated in either direction, with the answer probably being influenced by the size of the airport and its resources.

The five airports have different approaches to financial contingency planning. Some have written plans and operating capital requirements, while others have unwritten policies and procedures. In general, the five airports seek to have cash reserves equal to one to 2 years of operating costs no matter what the type of disruption. One airport has a very large line of credit, and one is seeking a line of credit. Several airports have very high limit credit cards for senior managers to use in case of emergency response or recovery procurements.

Interpretations from the Literature Review

The literature review for ACRP 03-68 was challenging, producing an unexpected paucity of plan integration artifacts. One interpretation is that the dearth of usable documents and literature, itself, is prima facie evidence that contingency plan integration is poorly understood and minimally practiced in the field. We see more evidence of plan "overlapping" than we do plan "integration," perhaps because various contingency plans are triggered across the incident timeline at different times. It may be an indicator that plan integration has not been widely considered, that it is a bad idea, or that it begs for additional analysis and evaluation through studies like ACRP 03-68. Because our literature review will be completed in October, we may discover additional evidence of integration that we have not observed to date.

Key Findings Narrative

Plan Integration Is Not Widely Seen as Beneficial

No one we interviewed initially perceived the need for integrating contingency plans. They mused about why integration might be beneficial, posing more detriments to us and identified more obstacles than they did tangible positives.

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Tool Parameters

In every case study these parameters for plan integration tools were seen as important:

- Short/brief, easy to use, easy to learn, easy to implement, and accessible, so they provide value for time-constrained line response and recovery managers.
- Accretive, not additive; cannot add a burden; must be de-conflicted and reference other plans.
- Action-oriented; focused on integration of the operational/behavioral dimensions, not the planning dimension; tactical, not strategic.
- Operational, not financial, and not FSPs.
- Considers city, county government partners and helps improve communication, coordination.
- Flexible, customizable, and scalable for applicability from small to medium-sized airports.
- Free.
- Tangible, not electronic.

Two "Flavors" of Contingencies

A theme running throughout our conversations was that airport plans for "recurring" or "standard expected" events may require a different approach to integration than plans addressing rare events. The former become internalized and operationalized as institutional knowledge, while the latter may require tactical aids and "reminder" tools because staff experience with them is uncommon.

Action versus Policy

There was strong consensus that integrating "policy-level" plans (long narratives in binders, e.g.) has less value than integrating "action-level" plans. However, because airports may use the ICS, the National Incident Management System (NIMS) and its offshoots like California's SEMS, or the Emergency Support Functions defined by FEMA and DHS, there may not be a single construct under which to naturally integrate various contingency plans. What these systems do well – but differently - is provide a framework that might be useful in dealing with all types of contingencies in a similar process. It is critical that the plans being considered for integration are those that are "tactical" and "action" oriented plans rather than strategic and policy-level plans. Integration means plans should be deconflicted, cross-referenced, and coordinated, and should realistically support the tactical response process. Action-based plans include checklists, field operating guides (FOGs), standard operating procedures (SOPs)

Integration at the Tactical Level

Considerations around "action-level" integration may include systems in which collaboration naturally occurs, such as tabletop, functional and full-scale exercises; drills and simulations; and the Emergency Operations Center (EOC). Tools to help airports ensure that the right players and plan stakeholders are "in the room" may be beneficial. Frequently occurring contingencies seem

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to benefit most from these sorts of non-plan integration tools whereas written tools such as SOPs, FOGs, and tactical guides may be far more important and useful for rarer types of contingencies.

Financial versus Operational Resilience

All airports seem to understand the need for operational and financial resiliency, but financial resiliency was not on the agenda of interviewees. Post-pandemic, focus seems to be on all forms of resilience including financial (but by senior airport management, rather than "front line response" managers who own most operational contingency plans), as discussed above. Airport CFOs agreed that financial contingency plans have such different goals and time frames that they cannot productively be integrated into operational contingency plans.

Importance of Senior Management Sponsorship & Support

Every airport agreed that "educating up" to top management was key to sustaining funding for contingency planning and that articulating contingency planning in the context of "airport resilience" was key at sustaining a "place at the table" for decision making.

Facility Security Plans

When coordinating contingency plans by whatever method, some airports felt they should factor in their ASP and related documents. But because contingency plans need to omit Sensitive Secure Information (SSI) to the greatest extent possible, which limits extent of FSP integration, the research team concluded that FSPs should not be considered for integrating with other contingency plans.

Integrating Dynamics from City/County Plans

Because airports that are part of municipal/county government are both a "node" on the city's COOP or Mass Casualty or Pandemic Plan and have their own airport specific plans (AEP, FSP, IROPS, etc.), it challenges them both because it adds the number of agencies necessary for collaboration and may constrain the nature of that collaboration. The challenge in these circumstances is understanding (as articulated by airport interviewees) "who's plan are we in at any point in time?", "whose resources are needed most?", "whose resources are being tasked?", and "does the airport need resources form other stakeholder jurisdictions, or is the airport the resource they need?" Airports stressed the importance of either staffing an internal Emergency Manager or working with an established external City/County Emergency Manager. Integration tools must consider the city/county government relationship many airports must deal with and be flexible enough to accommodate public and authority airports and the constraints imposed on them from that circumstance.

After-Action Plans

Few airports do formal AARs and Improvement Plans (IP) for incidents or exercises. Even fewer have formal tracking of action on IP items. Tools might include a template for AAR/IP's, a checklist for performing hotwash and AAR/IP, or a template for IP action tracking of corrective actions after simulations, drills, TTXs, FXs, and FSX's.

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Paper versus Electronic

Few airports use subscription electronic tools beyond having plans available on laptops and tablets. They are budget constrained and electronic plans may be too expensive to acquire and maintain.
Most airports do not want an additional electronic system they have to procure or subscribe to, beyond the online tools they have adapted to integrate contingency responses (Everbridge, Microsoft Teams, c.p.). While paper may be more labor-intensive to keep updated and synchronized/deconflicted than digital plans, the research team concluded that we should develop hard copy tools only.

Culture of Collaboration in Practice

“Collaboration” (a behavioral notion that can be built into airport culture) is seen as important as “integration” (a technical concept). However, a culture of collaboration and resiliency may be hard to create and sustain. It seems to be in the operationalization of plans that “integration” occurs. This can happen through training and exercises: daily briefing, drills, snap drills, training, so-called “tactical worksheets,” exercises (tabletop, functional, full-scale), FEMA courses, and NIMS/ICS training. It also occurs through experiencing large-scale regional events that force response integration by default.

Outcomes and Benefits

Tools must not just add more work or create the burden of another document to be responsible for, but instead must provide some real and new insights and best practices for how response can be improved by approaching response integration more effectively, so airports realize the integration benefits.

Cost/Benefit Analysis

Because funding is always the issue for airports, the tools we develop in ACRP 03-68 must be free. Cost-benefit analysis will be challenging, based on the ambivalence of airports for plan integration and the difficulty in identifying measurable benefits. The case studies revealed that the cost may essentially be the time airports do not have, and the enthusiasm they will not display if the tools are not widely perceived as effective in improving airports’ day to day preparation, response, and recovery activities. Quantitative metrics (such as time to restore normal operations, lost income, cost to recovery, or damage to or enhancement of the airport’s reputation or brand based on the quality of response and recovery) are not readily measurable.

How Time Constraints Impact Tool Development

Because small airports are highly constrained by staff time availability and because employees may resist learning a system that may be different from what’s done today, tools developed in this project need to be simple, accessible, easy to learn, easy to implement, and capture/document personally held institutional knowledge.

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Tool Development Direction

Tools need to be scalable for complexity of incidents, complexity of airports, and complexity of community contexts (including mutual aid). While the research team will address tool analysis thoroughly in Task 5 in early November, from the literature review to date and the case studies a clearer picture is emerging around the type of tools that will need to be developed in ACRP 03-68.

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Cost/Benefit Analysis

Using Return on Investment and Resiliency Return on Investment for Preparedness

Evidence for Practice

This paper gives examples of computing the hard and soft benefits of investments in resiliency taken from recent research into law enforcement and airports. It makes the case that the difficult process of quantifying soft benefits is worthwhile as Return on investment (ROI) remains a key factor in senior management and governing body decisions on investment.

Introduction

ROI has long been accepted as a primary tool for decision-making for capital investments and even choices among competing operational budget lines. More recently resiliency return on investment (RROI) has been introduced for the special case of investments in redundancy and other measures to enhance resiliency of facilities and activities (Hall et al., 2017).

ROI is defined as the total benefits (in dollars) of an investment divided by the total cost of that investment (Phillips and Phillips, 2009). In most cases ROI is computed for a fixed number of years, typically the expected useful life of the capital project. For non-capital projects, an arbitrary time span is usually set. The length of such an arbitrary time span depends on the nature of the industry, and recently, more often on the rate of technological transformation affecting that industry.

RROI is no different than ROI except it is applied to investments to improve the resiliency of a facility or the operations of an organization. Resiliency measures may be durable for longer than other types of projects and their benefits could last over a longer time frame. The definition of resiliency has several dimensions. The classic definition of being able to absorb a disruption and bounce back to normal certainly applies (ASCE, 2006), but it has different implications when looked at short term as in a natural disaster, accident, or criminal action compared to looking at a long-term trend in technology, change in business model, or nature of threat. RROI can be computed in each of these cases but with vastly different degrees of difficulty depending on the time frame and the uncertainties of longer term trends.

One difficulty in applying ROI and RROI is the paucity of peer-reviewed papers in the field. Much of the literature exists as corporate white papers, sales pitches, conference presentations,

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and similar gray literature. This paper seeks to bring the significant ideas from some of that literature together in a useful and citable form.

This paper will focus on the extreme difficulties of applying ROI and RROI to things like law enforcement, security, emergency preparedness, and resiliency because they are states of being and often not so tangible or easily measurable. Examples from law enforcement, airport security, and airport emergency management will be used to illustrate the methods and challenges of developing meaningful ROI and RROI measures. The paper will also emphasize the key importance of clear communication and how using ROI and RROI can foster or obscure it.

Why Resiliency Matters

The main reason that resiliency matters is that we now live in an environment where disruptions are the normal state. Any non-resilient program or structure is prone to sudden or gradual failure if its leadership has not invested in resiliency measures. The disruptions may be relatively sudden like an earthquake, tornado, industrial accident, fire, ransomware attack, or terrorist incident or they may be slow such as changes in technology, markets, or business models. When lack of resiliency is viewed in terms of opportunity cost, where opportunity cost represents the benefits forgone by choosing one option over another (Sharkey, 2022), then non-readiness for a pandemic, a plane crash, a ransomware attack, or whatever could be negatively measured in millions of dollars, many lives, and diminishment of public confidence. Often the argument becomes about “black swans” (Taleb, 2007) where the old dilemma of not paying attention to risks that are so unlikely justifies not trying to measure “contingent ROI.” When the probability is really difficult to measure conventionally and the derived “cost” is so extreme, all evaluation of ROI is hard, but still worthwhile.

Applying ROI to Law Enforcement

Applying new technologies to law enforcement provides the best picture of the challenges and benefits of computing ROI. The usual sales pitch for new technology for law enforcement is that it will increase productivity or decrease staffing needs, or both. A more subtle argument would be that technology can increase security and safety. A simple computation of ROI would typically take the total money saved on salaries, benefits, and equipment and divide it by the cost of procuring the new technology; in other words, dividing the hard benefits by the costs to get ROI (Gascón and Foglesong, 2010). As traditionally applied in the private for-profit sector, “hard” ROI is defined as the ratio of money gained (profit) or lost, relative to the amount invested (total cost) (Tarantino, n.d.). However, in government and perhaps especially in public safety functions such as law enforcement, looking only at hard benefits may exclude soft benefits that exceed or even greatly exceed the hard benefits (L-tron, n.d.). Nevertheless, Tarantino notes that “hard” ROI measures are great tools and can serve as a quantifiable backdrop to the other,

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"softer" measures that we will examine next. Tarantino (n.d.) also notes that "hard" ROI sometimes fails and has unintended consequences. An example of this occurs when ROI is used to justify a reduction of personnel, which can impact law enforcement operations, creating staffing challenges and loss of public confidence that often outweigh cost savings.

What are hard benefits and what are soft benefits? Hard benefits are things that are easy to cost. Examples are savings with identifiable costs such as reduction in force or decreased operating costs of equipment. Soft benefits usually fall in the realms of political, social, ideological, strategic, and community stewardship benefits (Tarantino, n.d.).

Tarantino (n.d.) sorts soft benefits to consider for law enforcement into three categories: stewardship, time, and personnel. Sample soft benefits that he cites include improved ability to perform mission-critical functions and meet public safety objectives; reduced errors, rework, and duplication of effort; increased job satisfaction; and reduced administrative burden on staff. In addition, increased retention and enhanced recruitment are soft benefits (QuinnWilliams, 2022). The main problem with soft benefits is how to calculate their dollar value. Tarantino argues that they can be broken down into measurable components such as overtime costs, training costs, recruiting costs, etc. This analysis is both time-consuming and intellectually challenging, but when done, it shows the "hard" roots of the "soft" components of ROI, thereby allowing a more complete picture of the true benefits for a given investment in law enforcement.

Let's look at applying ROI to a major area of innovation in law enforcement: the use of cameras including CCTV area surveillance, dashboard cameras, body cameras, traffic cameras, and the many types of integrative software that allow such things as facial recognition, license plate reading, tracking of individuals, and perimeter security, to name just four applications of modern camera systems. The traditional ROI argument would be that the increased areal and temporal coverage allowed by such systems as compared to patrolling police would generate significant cost savings primarily through a reduction in the number of patrol police needed along with their equipment. However, the camera systems detect more incidents that require a human to investigate, which may actually increase the number of officers required. Where camera systems do generate a time savings is on the investigatory end of incidents. An example of a soft benefit of the use of cameras in law enforcement is greater transparency.

Applying ROI to Airport Security

At the recent Passenger Terminal Expo and Conference, one of the major focuses was on the application of artificial intelligence and machine learning to passenger and baggage screening in airport checkpoints (Passenger Terminal Expo and Conference, 2023). Nearly every presenter and panelist from an airport (e.g., Evans, 2023; Dullard, 2023) stated the importance of showing a positive ROI to get approval for these very expensive new technologies.

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Another area of focus at the same conference was the use of advance imaging technology to eliminate the limitations on liquids, aerosols, and gels in carry-on baggage going through security checkpoints. And again, nearly every speaker (e.g., Reiter, 2023) explicitly stated the need to use ROI to convince airport senior management to invest in the equipment, space to house it, and training of operators to use it effectively. Furthermore, the national and international civil aviation authorities and transportations security agencies will also demand ROI information as they consider revising regulations.

Applying ROI to Airport Emergency Preparedness and Resiliency

Airport emergency preparedness and resiliency include many types of activities: for example, staffing, training, drilling, exercises, facility design, facility construction techniques, redundancies, and back-up power (Hanson Professional Services, in progress) and communications systems. These factors all add cost above the bare minima for creating an airport that can function under normal conditions, so ROI and RROI come into play for justifying the additional costs to achieve preparedness and resiliency.

Let's examine three types of resiliency activities and how ROI and RROI apply to them: (1) the personnel issues around staffing, training, drilling, and exercising, (2) operational redundancy, and (3) facility hardening and redundancy.

Personnel Issues Involved in Airport Resiliency

Over the past 20 years a gradual trend has emerged for airports to add professional emergency managers to their staffs. Part of this was a response to 9/11 and to the major hurricanes of 2004 and 2005, and part of this was a response to increasing pressure from the FAA to improve airport emergency planning (FAA, 2009). This trend has run in parallel with the development of emergency management as a professional field. As a result, the number of U. S. airports with full-time emergency managers has grown from about five in 2004 to more than 110 in 2022. This is out of approximately 550 U.S. airports that have commercial service. In addition, many of the larger general aviation airports have emergency managers, either full-time or as a collateral duty. In parallel with the trend to add emergency managers to airports' staffs has been a trend to add training and exercise specialists. Often the emergency manager serves both roles. Training is focused on both emergency response roles and on airport security. Typically, training occurs during employee onboarding and then at various times during the year. Training and drills tend to be synched to the FAA's requirements for annual and triennial emergency exercises and TSA's requirement for an annual aviation security exercise. Some airports do far more training and exercises than these minima (Smith, Garcia, Sawyer and Kenville, 2016).

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It is easy to cost out adding positions for emergency management and training, and the costs of doing training and exercises are relatively small, with the major cost issue associated with training and exercises being time away from other activities of airport employees and those of other stakeholders. The hard benefits of these expenditures are more difficult. How much is being able to check the box for required exercises worth? How much is having zero discrepancies on inspections worth? In most cases the soft benefits greatly outweigh the hard benefits. What is a preparedness culture worth at an airport if a very rare event like an active shooter, plane crash, or pandemic happens? What other soft benefits accrue to an active training and exercise program? Bersin (2010), looking at highly successful for-profit corporations, found that the more active a firm's "learning and development" program for resiliency is, the greater are the firm's profits, growth, employee retention, and internal capabilities. Usual measures of ROI don't capture these greater benefits. Airports can logically be expected to gain similar soft benefits except rather than greater profits, airports would free up funds for other programs.

Operational Redundancy

A straightforward example of operational resiliency is the provision of back-up electrical power. Severe weather events and regional electrical grid failures have shown airports that they need to have on-site capabilities for generating electricity. These needs come in several levels: the minimum to run airfield lighting, avionics, and air traffic control to support aircraft operations; the minimum for life safety; the minimum to evacuate a terminal; the minimum to operate the entire airport. The cost of back-up power escalates greatly across these four tiers, but the ROI and RROI computations are similar. Procurement, installation, maintenance, training to operate, and fuel storage are easy to cost out. Depending on which tier is being examined for ROI, the costs of an airport being shut down are also relatively easy to calculate. Here the hard benefit is how much cost is avoided by keeping the airport operational or by minimizing the time it is shut down.

The soft benefits of back-up power mostly accrue in the stewardship dimension. The airport benefits in terms of its reputation and brand. Perhaps the greatest benefit of airport resiliency is its ability to stay in operation or return to operation quickly so as to be able to be a major asset in local or regional disaster response and recovery (Smith, Kenville and Sawyer, 2015).

Facility Hardening

Designing and building airport facilities to be resilient increases their cost, sometimes significantly. The most economical approach is to design resilient features into new construction whereas retrofitting existing buildings is more costly, difficult, and disruptive of ongoing operations. An example of this is the hardening of Memphis International Airport against damage from a potential major earthquake from the New Madrid Fault (Thomas Wallace,

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interview, July 28, 2022). The airport has been carrying out a multi-year program of seismic upgrades of runways, taxiways, bridges, the terminal, and new construction of a concourse. Resiliency is particularly important at Memphis' airport because of its central role in logistical supply chains nationally and worldwide and its potential role in post-earthquake disaster response and recovery of its region when most of the highway and rail infrastructure may be destroyed or severely damaged. Taking into account the direct and indirect benefits of the airport in this role is challenging but not impossible if each of the dimensions in Tarantino's (n.d.) model is broken down into functional components and costed individually.

Advantages and Disadvantages of Using ROI and RROI

The main advantage of computing ROI and RROI for resiliency is to make a case for investment in terms that senior management understands and can use to compare different programs demanding funding. ROI is as near to being a universally accepted decision-making tool as there is. Perhaps the main advantage of using the RROI concept is to emphasize inclusion of "soft" benefits and perhaps also longer term costs and benefits.

The main disadvantage for computing ROI and RROI for resiliency investments is how time-consuming and intellectually demanding it is to capture the hard benefit subcomponents of soft benefits. Using these types of ROI measures may require an adjustment of worldview or culture on the part of decision-makers. Fortunately, a culture of resiliency as well as of sustainability seems to be emerging as a new generation of leaders takes over.

Conclusions

It is tempting to argue that ROI/RROI is immaterial to investment decisions given that the price of most resiliency measures is small or even tiny compared to the costs of non-resiliency when something happens. Although this argument is most obvious when applied to very rare events with huge consequences—the black swans—it pertains to all investments in resiliency, preparedness, and prevention. The United Nations Disaster Programme (2016) estimates "that for every dollar spent reducing disaster risk, seven dollars will be saved from economic losses associated with cleanup and recovery." ROI and RROI are worth computing although the effort to include all soft benefits and actual costs can be onerous and time-consuming.

Normal budgeting processes, whether for capital or operating expenditures, usually involve competition among various programs. Senior management will typically use ROI as a deciding factor, perhaps even the most determinant factor, in making funding decisions. For this reason, entities seeking to improve their resiliency must be prepared to do the work to compute ROI and RROI.

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Investments in resiliency are hard to justify because they are bets on a future that decrease the odds of seeing a return. In other words, the greater the investment in resiliency, the less likely resiliency will be needed and the harder it becomes to justify it was necessary in the first place. If "there is no glory in prevention" then RROI calculations only make sense for measures reducing (mitigating) but not preventing the impact of an event. It's a disconcerting thought that resiliency actions may have to plan for some damage to occur so that the investment can be better evaluated and justified in hindsight (Daniel Link, email, November 28, 2022).

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Smith, James F., Ricardo E. Garcia, John M. Sawyer, and Kimberly A. Kenville. 2016. *Tabletop and Full-Scale Emergency Exercises for General Aviation, Non-Hub, and Small Hub Airports*. Airport Cooperative Research Program (ACRP) Synthesis of Practice 72. Washington, DC: Transportation Research Board. Accessed September 28, 2022. https://www.trb.org/Publications/Blurbs/174692.aspx

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Literature Review

General Search Terms:[1]

"Enterprise risk management" OR ERM
Airport AND contingency AND plan*
Airport AND emergency AND plan*
Airport AND "business continuity" AND plan*
Airport AND continuity AND operation* AND plan*
Airport AND cybersecurity AND plan*
Airport AND cyber* AND plan*
Airport AND financial AND sustainability AND plan*
Airport AND (resiliency OR resilience) AND plan*
Airport AND recovery AND plan*
Airport AND "hazard mitigation" AND plan*
Airport AND crisis AND (management OR communications) AND plan*
(Integrated OR integrating OR integration) AND contingency AND plan* [general, not just airports]
Contingency AND plan* AND (("critical infrastructure" OR "key resource") OR "CI/KR")
(Electronic OR online) AND emergency AND management AND tool

Non-Airport Case Studies:

"Fusion center" AND Nevada*
Hazard mitigation AND plan AND Nevada*
Hazard mitigation AND plan and Missouri Southern State University OR MSSU
Hazard mitigation AND plan AND "Port of Vancouver"
"Comprehensive emergency management" AND plan AND Nevada
"Comprehensive emergency management" AND plan AND "Missouri Southern State University" OR MSSU
"Comprehensive emergency management" AND plan AND "Port of Vancouver"
All general terms above, substituting "Nevada", "State of Nevada", "Missouri Southern State University", "MSSU", and "Port of Vancouver" for "Airport"

Additional Search Terms:

Integrating: "integrated, integration, comprehensive, combined, coordinated, deconflicted"
Contingency: "contingencies, emergency, operational, recovery, response, crisis, cyber, disaster"
Plans: "planning"

1 In our Boolean search process, an asterisk allows any form of the base word, e.g., plan, plans, and planning.

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Literature Review Introduction

The research team struggled with how to prove something does not exist. In the ACRP 03-68 literature review, we used citation analysis of key documents and authors as well as Boolean searches across databases, industry publications, and proceedings to test the hypothesis and commonly held belief, revealed in our case study interviews, that there is no acknowledged value in contingency plan integration. This is reflected in the annotations to sources which discuss citation patterns of these influential publications and authors.

The literature review also proved the negative by showing the demonstrable positive interest in other threat mitigation and contingency planning resources for airports. A 2005 wildlife hazard management manual by Edward C. Cleary & Richard A. Dolbeer has been cited 199 times. Georgia Lykou, Argiro Anagnostopoulou, & Dimitris Gritzalis's 2019 paper on cybersecurity threat mitigation for small airports has already been cited 58 times while a 2020 paper by the same team on UAS threat mitigation has been cited 91 times. If there is a demonstrable need, airports and airport scholars will find and use these resources.

In other cases, in this literature review, we saw where resources that deal with airports get adopted and cited outside of the airport community but not within it. Fr example, Colin B. Gabler, Robert Glenn Richey Jr., & Geoffrey T. Stewart's 2019 article discusses airports and other critical infrastructure and key resources in context of public-private collaboration in disaster resilience. The article cites 52 sources and is cited by 65; none of these citations in either direction primarily focus on airports.

Integrated Contingency Planning and Crisis Communication Plans

de Fátima Silva Piedade, Lúcia & Jorge Silva. "Optimized Communication Plan and Its Impact on the Contingency Plan Previously Put in Place for Timely Crisis Response in the Air Sector." *VII RIDITA - International Congress of the Iberoamerican Air Transportation Research Society*. V.1, 465-473. Covilhã, Portugal: University of Beira Interior, October 11, 2019. https://ubibliorum.ubi.pt/bitstream/10400.6/9643/1/VII-RIDITA-LPiedade.pdf

> Looks at the contingency planning, specifically the crisis communication and IROPS plans for the Lisbon airport, through the lens of faster response times to a range of typologies of crisis. The only airport specific sources in their literature review are the crisis communication and IROPS plans for the studied airport. Only an outside industry paper (oil & gas) cites this work.

de Fátima Silva Piedade, Lúcia & Warnock-Smith, David. "Crisis and Communications Management in the Airports Sector: The Effective Use of Social Media." *Journal of Air Transport Studies*. 12, no. 1. 1-24. January 1, 2021. https://doi.org/10.38008/jats.v12i1.173

> "Using an airport user questionnaire (in Portugal) and operational expert interviews (in the UK), this study aimed to determine the ways in which airport use of social media platforms should be integrated into overall airport crisis and disruption management strategies."

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Fischer, Robert J., Edward P. Halibozek, & David C. Walters. "Contingency Planning Emergency Response and Safety." *Introduction to Security.* 10th edition. 249-268. 2019. doi: 10.1016/B978-0-12-805310-2.00011-1

> Discusses the fundamental elements and need for effective integrated contingency planning for the four major components of contingency planning: emergency response, crisis management, business recovery, and business resumption. Cited by 10 publications across hospitality, healthcare, education, petrochemical, and small business fields, none in airport field.

Gabler, Colin B., Robert Glenn Richey Jr., & Geoffrey T. Stewart. "Disaster Resilience through Public-Private Short-Term Collaboration." *Journal of Business Logistics.* 38, no. 2. 130-144. March 2, 2017.
https://doi.org/10.1111/jbl.12152

> Discusses airports and other critical infrastructure and key resources in context of public-private collaboration in disaster resilience. Cites 52 sources and is cited by 65; none of these citations in either direction primarily focus on airports.

Grigsby Smith, Deborah. "Centennial Airport: A Case Study for Small Airport Social Media Strategy." *Journal of Airport Management.* 13, no. 3. 225-237. June 1, 2019.

> "Large commercial airports dedicate robust resources to marketing and communication efforts: investing heavily in dedicated staff and consulting firms to build community, increasing stakeholder engagement and aggressively managing reputations during times of crisis. Smaller regional, non-hub and general aviation airports face many of the same challenges, but often with fewer available resources and resulting in greater image vulnerability." Discusses Centennial's crisis communication plan, first written in 2017.

Mantzana, Vasiliki, Eftichia Georgiou, Ioannis Chasiotis, Ilias Gkotsis, Tim H. Stelkens-Kobsch, Vasileios Kazoukas, Nikolaos Papagiannopoulos, Anastasios Nikas, & Filippos Komninos. "Airports' Crisis Management Processes and Stakeholders Involved." *Annals of Disaster Risk Sciences: ADRS.* 3, no. 1. 2020.

> "Any physical or cyber incident that causes loss of infrastructure or massive patient surge, such as natural disasters, terrorist acts, or chemical, biological, radiological, nuclear, or explosive hazards could affect the airports' services provision and could cause overwhelming pressure. During the crisis management, several stakeholders that have different needs and requirements, get involved in the process, trying to cooperate, respond and support recovery and impact mitigation. The aim of this paper is to present a holistic security agenda that defines the stakeholders involved in the respective processes followed during the crisis management cycle. This agenda is based both on normative literature, such as relevant standards, guidelines, and practices and on knowledge and feedback extrapolated from a case study conducted in the context of the SATIE project (H2020-GA832969). In meeting paper's aim, initially the normative review of the phases of the

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crisis management cycle (preparedness, response, recovery and mitigation) in the context of airports as well as general practices applied, are presented. Moreover, the key airport stakeholders and operation centres involved in airports operations, as well as during the crisis management are analysed. By combining the information collected, a holistic cyber and physical crisis management cycle including the stakeholders and the relevant processes are proposed."

Quarshie, Anne M. & Rudolf Leuschner. "Interorganizational Interaction in Disaster Response Networks: A Government Perspective." *Journal of Supply Chain Management.* 56, no. 3. 3-25. https://doi.org/10.1111/jscm.12225

Uses a qualitative research approach to study New Jersey state government's response to Hurricane Sandy in interaction with other humanitarian actors and interorganizational interaction as a whole. Cites 17 sources, cited by 23, none of which are primarily airport focused.

Enterprise Risk Management

Gatt, Marnick, Simon Grima, & Yannis E. Thalassinos. "An Enterprise Risk Management (ERM) Maturity Index for European Airports." International Conference on Computational Methods in Enterprise Economics. *CMEE 2019: Decision Making in Management*. 337-378. August 11, 2021.

"Structured interviews with ten airport risk management practitioners and consultants were conducted. Data generated from these interviews attested to twenty-two applicable ERM best practices, which were divided into eight categories…Results show senior management commitment, an optimum communication process, and well defined ERM roles and responsibilities. These are the three most crucial factors for airports to achieve a high degree of ERM maturity."

Messer, Ron. "Risky Business: Using Enterprise Risk Management at Airports." *Journal of Airport Management*. 11, no. 2. 202-213. March 1, 2017.

"The most commonly identified concern related to major capital projects, while the activities of the airlines were also frequently mentioned. Of the 220 risks noted, 74 were considered to be above the risk tolerance threshold. The information in this paper can be used by airport operators to help identify, assess and mitigate the potentially negative effects of threats in their internal and external environment." Ron Messer worked at the Vancouver Airport Authority for 11 years, including as the Head of Audit Services before serving as faculty at Kwantlen Polytechnic University. This paper has been cited twice, neither time by fellow airport researchers.

While Messer's general budgeting articles are cited more (the most is cited 17 times), his airport specific ones on airport land development, REITs, and missing revenues are cited 0-3 times.

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National Academies of Sciences, Engineering, and Medicine. *Airport Cooperative Research Program Report 74: Application of Enterprise Risk Management at Airports*. Washington, D.C.: The National Academies Press, 2012. https://doi.org/10.17226/22744.

Only one documented citation- Gatt et al. article listed above.

Snyder, Paul, Gary Ullrich, & Zachary Waller. *Airport Cooperative Research Program Synthesis 106: Airport Risk Identification and Prioritization Practices.* Washington, D.C.: Transportation Research Board of the National Academies, 2020.

Airport Business Continuity Planning and Financial Sustainability Planning

Adetula, Akinsote Smith. *A Quantitative Descriptive Study of Business Continuity Management Practice of the Airports in the District of Columbia, Maryland, and Virginia States.* PhD thesis. Colorado Springs, CO: Colorado Technical University, 2021.

Cites both Corzine 2013 and Camastral 2014.

Camastral, Marisa Rita. *Business Continuity Management in Airports: Securing Continuity in the Face of Crisis.* PhD thesis. Brisbane, Australia: Queensland University of Technology, 2014. https://eprints.qut.edu.au/72795/1/Marisa_Camastral_Thesis.pdf

Camastral identifies in the literature review of her 2014 dissertation the lack of guidance on BCM implementation in complex infrastructures like airports and takes as a central research question optimal approaches from BCM implementation at a range of airports. Camastral 2014 has been downloaded 1895 times since publication including 240 times in the past 12 months. It has been cited 3 times, mostly within the airport field: a 2016 paper on assessing safety and security disruptions in Australian airports; a 2019 paper on BCM framework for disaster resilient SMEs in Malaysia; and a 2018 paper on the noise zoning plan for Lodz Airport in Poland.

Corzine, Scott. *Airport Cooperative Research Program Report 93: Operational and Business Continuity Planning for Prolonged Airport Disruptions.* Washington, D.C.: Transportation Research Board of the National Academies, 2013.

Cited by Adetula 2021 as well as a 2016 thesis on safety and security disruptions at Australian airports and ranking building automation and control system (BACS) components on their business continuity impact.

Mansilla, Patricio & José Manuel Vassallo. "Innovative Infrastructure Fund to Ensure the Financial Sustainability of PPP Projects: The Case of Chile." *Sustainability.* 12, no. 23. 2020. https://doi.org/10.3390/su12239965

Discusses airports in the context of a broader discussion of public private partnership (PPP) best practices and financial sustainability. Cited by 7 publications, including a 2021 working paper on resilient infrastructure in Indonesia which discusses airports.

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Romig, Thomas. "A Collaborative Approach to Business Continuity and Organisational Resilience when Dealing with a Major Crisis, such as COVID-19." *Journal of Airport Management*. 15, no. 3. 226-234. June 1, 2021.

> Romig is the vice president of safety and operations at Airports Council International. "This paper explores some of the key aspects of tackling business continuity planning from an airport system approach, dealing not only with operational resilience but also with economic sustainability, and provides a case study of the approach taken at Geneva Airport in Switzerland."

Tidhar, Roni. "Airport Emergency Management: The Art of Strict Rules and Flexible Minds." *Journal of Business Continuity & Emergency Planning.* 13, no. 3. 230-239. Spring 2020.

> Tidhar uses airports as a best practice model for maintaining operational continuity in the face of a crisis. "This paper suggests that airports provide a good case study from which to learn, due to their emergency-oriented nature; their nonstop tango with disruptions of many origins; their experience in moving from crisis to normal operations as quickly as possible; and because of their multifaceted functions and stakeholders." This article has been cited once by a publication outside the airport field.

Continuity of Operations Planning

Echeverria, Kelly, Jessica Adams, Duane Lohn, & Jennifer Posten. "Continuity of Operations Planning." Presented at Nevada Emergency Preparedness Association's Nevada Preparedness Summit 2022. February 2022. Slides 91-115. https://preparenv.org/resources/Documents/NPS%202022%20Slide%20Deck.pdf

> Washoe County Emergency Manager Kelly Echeverria and representatives from vendor RSI discuss NevadaContinuity.com. Reno-Tahoe (RNO) is one of the organizations currently utilizing NevadaContinuity.com.
>
> The work of Kelly Echeverria and her team has been highlighted by FEMA (see U.S. Department of Homeland Security 2018) as a best practice for continuity of operations planning at a statewide level. Echeverria presents widely on the topic within Nevada and the industry. As of November 2022, I could not find any academic publications on Echeverria and her team's work.

U.S. Department of Homeland Security. Federal Emergency Management Agency. "Continuity Guidance Circular." February 2018. https://www.fema.gov/sites/default/files/2020-07/Continuity-Guidance-Circular_031218.pdf.

> In 2009, the State of Nevada began Continuity of Operations Planning. This statewide collaborative work involving more than 25 jurisdictions was highlighted in this FEMA Continuity Guidance Circular. From page 34: "The State of Nevada and localities, tribes, and other stakeholders developed a multi-year initiative to develop and update continuity of

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operations and continuity of government plans and successfully used Homeland Security Grant Program (HSGP) funds to assist implementation.

"The Nevada State Homeland Security Strategy identified continuity of operations and continuity of government planning as one of the top three areas of focus when the initiative began. Nevada State law further governs continuity planning within the State by requiring "plans for the continuity of the operations and services of the political subdivision, which plans must be consistent with the provisions of Nevada Revised Statute 239c.260."

"In an effort to increase continuity capabilities statewide, a project team developed a multi-year project plan to enhance continuity planning and integrate into existing efforts throughout the State and sustain these efforts in future years. The Nevada Commission on Homeland Security approved the use of HSGP funds to support these efforts. The team developed a continuity task force, comprised of state, local, and tribal government representatives, volunteer agencies, and the private sector, to serve as a steering committee for the effort.

"In the first year of the project, the team focused on continuity planning efforts for six counties in the northern part of the state. Subsequent grant cycles enabled extension of the planning effort to the remaining southern and eastern counties while sustaining previous efforts. The HSGP funding used in support of this effort funded task force meetings, planning efforts, and a series of training workshops and resulted in viable continuity plans and templates for jurisdictions throughout the state. In addition, by coordinating participation throughout the State under the limited timeframe of grant funding, the project enabled a cooperative and collaborative approach to continuity planning that leveraged best practices, lessons learned, and shared experiences."

Nevada COOP Planning. https://www.nvcoop.org/

Project managed by Washoe County's Emergency Management team, Nevada COOP provides COOP training and planning support. This catchall public facing site highlights the two current Planning Grant Initiative focus institutions (two colleges) with links for trainings.

Airport Emergency Planning

Murphy, Stephanie, Ashlee Herring-Delventhal, Crystal Kline, & Blanca Rand. *ACRP Synthesis 115: Practices in Airport Emergency Plans.* National Academy of Sciences, Transportation Review Board. 2021. https://dx.doi.org/10.17226/26077

Page 47-51 of ACRP Synthesis 115 is a case study of LAS highlighting the innovative way then-McCarran socializes their Airport Emergency Plan with stakeholders. Appendix E includes a sample quick reference guide LAS developed. "Prior to 2017, all plans and exercises were additional duties divided among personnel, including the operations coordinator, who later became the airport's first emergency administrator. He felt the AEP was a good start, but just a start. While it complied with FAA 14 CFR, part 139,

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regulations, he realized the AEP is too large of a document to be physically used during an emergency. Knowing the AEP contains valuable information, he consolidated several checklists from information within each AEP annex for the various stakeholders that would be involved in an incident. In doing so, it became clear that there were some gaps in planning that the AEP did not address. For example, the AEP does not address incidents that take place inside the terminal, such as an active shooter event or the discovery of luggage with a large amount of illegal drugs. Another issue was nonstandardized terminology. Although the NIMS was fully adopted by LAS in accordance with the FAA's AC 150/5200-31C, there can still be residual nomenclatures that persist throughout an airport or aviation environment. For example, an airline may use the term "friend and family assistance center," instead of the NTSB's term, "family assistance center."

Hazard Mitigation and Planning

Brown, Stephanie & Brian Dillman. "Utilizing Organization Change Management Tools to Implement Safety Management Systems (SMS) Principles." *Collegiate Aviation Review International*. 39, no. 2. December 10, 2021.

Carson City, Nevada. "Carson City Hazard Mitigation Plan." July 14, 2021. https://www.carson.org/home/showpublisheddocument/77333/637668615401870000

Discusses CXP Master Plan and development, severe weather events at RNO, and a mercury spill leading to the evacuation of an apartment complex on Airport Road.

Chacko, Josey. *Sustainability in Disaster Operations Management and Planning: An Operations Management Perspective.* PhD thesis. Blacksburg, VA: Virginia Polytechnic Institute and State University, 2014. https://vtechworks.lib.vt.edu/bitstream/handle/10919/71759/Chacko_J_D_2015.pdf?sequence=1

While not airport specific, provides a good overview of multi-hazard management and planning.

Cleary, Edward C. & Richard A. Dolbeer. "Wildlife Hazard Management at Airports: A Manual for Airport Personnel." USDA National Wildlife Research Center - Staff Publications. 133. 2005.

This 2005 manual on wildlife hazard management has been cited 199 times.

Lykou, Georgia, Argiro Anagnostopoulou, & Dimitris Gritzalis. "Small Airport Cybersecurity: Threat Mitigation and Cyber Resilience Controls." *Sensors.* 19, no. 1. 2019. https://doi.org/10.3390/s19010019

"Smart airports strive to provide optimal services in a reliable and sustainable manner, by working around the domains of growth, efficiency, safety and security. This article researches: (a) the implementation rate of cybersecurity measures in commercial airports; (b) malicious threats that evolve due to IoT and smart devices installed; (c) risk scenario analysis for IoT malicious attacks with threat mitigation actions. With the aim to enhance

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operational practices and develop robust cybersecurity governance in smart airports, we present a systematic and comprehensive analysis of malicious attacks in smart airports, to facilitate airport community comprehend risks and proactively act, by implementing cybersecurity best practices and resilience measures." Cited by 58 publications, demonstrating a clear interest in this from small airports. One 2020 publication by the same authors, "Defending Airports from UAS: A Survey of Cyber-Attacks and Counter-Drone Sensing Technologies" has in turn by cited by 91 publications.

U.S. Department of Homeland Security. Federal Emergency Management Agency. "Local Mitigation Planning Handbook." March 2013. https://www.fema.gov/sites/default/files/2020-06/fema-local-mitigation-planning-handbook_03-2013.pdf

Washoe County Emergency Management and Homeland Security. "Washoe County Regional Hazard Mitigation Plan." Last updated 2020. https://dem.nv.gov/uploadedFiles/demnvgov/content/About/Washoe_Regional_HMP_2020.pdf

Excellent All Hazards mitigation plan that heavily involves the 4 publicly operated airports in the county, Reno-Tahoe International Airport (RNO), Reno-Stead Airport (RTS), and Spanish Springs Airport (N86) in Reno, and the Empire Airport (1A8) in Empire. Hazards discussed vis-à-vis county airports include hazardous materials, severe storms, and aircraft incidents.

Also discusses past small aircraft crashes in Washoe County are associated with small aircraft:

- August 30, 2016: A single-engine Beechcraft A36 crashed into the River's Edge RV Park in Sparks. The pilot and one passenger were killed. The plane did not directly hit any RVs; however, a fire started by the crash spread into surrounding RVs and vehicles. Residents were evacuated following the crash but were allowed to return home once conditions were safe.
- September 16, 2011: The Galloping Ghost, a highly modified North American P-51D Mustang racing aircraft, crashed into spectators while competing at the Reno Air Races, killing the pilot and 10 people on the ground. An additional 69 people on the ground were injured. This was the third-deadliest airshow disaster in U.S. history, following accidents in 1972 in Sacramento, California, and 1951 in Flagler, Colorado.
- January 21, 1985: Galaxy Airlines Flight 203. Shortly after takeoff from Reno-Cannon International Airport (now Reno-Tahoe International Airport), the Lockheed L-188 Electra four-engine turboprop crashed about 1.5 miles from the end of the runway and burst into flames, killing all but one of the 71 passengers onboard.

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